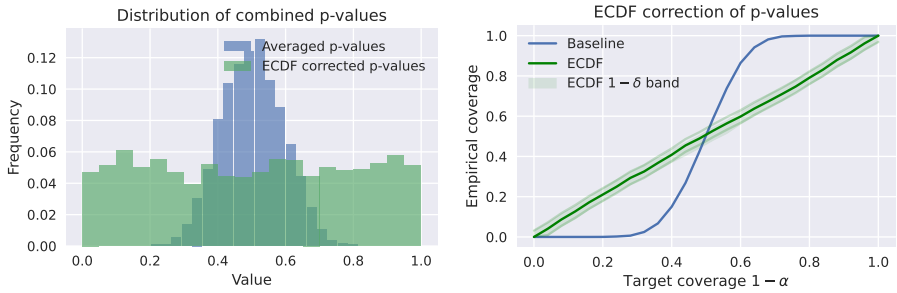

Figure 7: For illustration, we sample 10k p-values for $m = 10$ independent tests and duplicate them to obtain $m = 20$ dependent tests. Left: Histograms for averaged and ECDF-corrected p-values. The ECDF correction is able to ensure that p-values are distributed approximately uniformly. Right: Target confidence level α plotted against the empirical confidence level when calibrating with averaged p-values (blue) and the ECDF-corrected ones (green). The Dvoretzky–Kiefer–Wolfowitz inequality provides tight finite-sample guarantees on the ECDF correction, using here $\delta = 0.0001 = 0.01\%$.

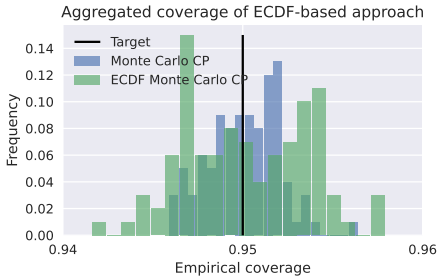

Figure 8: On our toy dataset, we plot aggregated coverage obtained using Monte Carlo CP without and with ECDF correction for $m = 10$ and $l = \lfloor n/2 \rfloor$. As expected, Monte Carlo CP, even for $m \geq 2$, does not result in reduced coverage. Thus, ECDF correction does not yield meaningfully different results besides exhibiting higher variation in coverage across calibration/test splits due to the additional split l.

2006)) provides rigorous finite sample guarantees for the ECDF, i.e.

$$\mathbb{P}(\sup_{f \in [0,1]} |\bar{F}(f) - F(f)| > \epsilon) \leq 2\exp(-2(n-l)\epsilon^2). \tag{14}$$

Basic manipulation results in a confidence band for $\bar{F}$

$$\mathbb{P}(\forall f \in [0,1], \bar{F}_-(f) \leq F(f) \leq \bar{F}_+(f)) \geq 1 - \delta \quad \text{with} \quad \bar{F}_\pm(f) = \frac{\min}{\max}\left\{\bar{F}(f) \pm \sqrt{\frac{1}{2(n-l)}\log\frac{2}{\delta}}, \frac{1}{0}\right\}. \tag{15}$$

This confidence band is illustrated shown in Figure 7 (right). We can now define the prediction set $C(X) = \{y \in [K] : \rho_y > \alpha\}$ for $\rho_Y = \bar{F}_+(\bar{\rho}_Y)$ and show that it satisfies $\mathbb{P}_{\text{agg}}(Y \in C(X)) \geq (1-\alpha)(1-\delta)$. Indeed, writing $\bar{F}_+ \geq F$ to abbreviate $\bar{F}_+(f) \geq F(f)\ \forall f$, we have $\mathbb{P}(\bar{F}_+(\bar{\rho}_Y) > \alpha) \geq \mathbb{P}(\bar{F}_+(\bar{\rho}_Y) > \alpha | \bar{F}_+ \geq F)\mathbb{P}(\bar{F}_+ \geq F) \geq \mathbb{P}(F(\bar{\rho}_Y) > \alpha | \bar{F}_+ \geq F)\mathbb{P}(\bar{F}_+ \geq F)$. Now we have from Equation (14) that $\mathbb{P}(\bar{F}_+ \geq F) \geq 1 - \delta$ and $\mathbb{P}(F(\bar{\rho}_Y) > \alpha | \bar{F}_+ \geq F) = \mathbb{P}(F(\bar{\rho}_Y) > \alpha)$ as the probability in Equation (14) is over $(X_i, Y_i)_{i \in \{l+1,\ldots,n\}}$ while $\bar{\rho}_Y$ is only a function of $(X_i, Y_i^j)_{i \in [l], j \in [m]}$ and (X, Y). Using the fact that $\mathbb{P}(F(\bar{\rho}_Y) > \alpha) \geq 1 - \alpha$, we obtain the desired coverage guarantee.

Figure 8 compares the empirical coverage obtained using standard CP with true labels to both approaches of Monte Carlo CP (cf. Algorithms 1 and 2) to verify that this approach indeed obtains coverage $1 - \alpha \approx (1-\alpha)(1-\delta)$ on average for $\delta \ll 1$. As discussed, without the ECDF correction discussed here, this is only empirical as the approach discussed in the previous section can only guarantee $1 - 2\alpha$. We also did not observe a significant difference in prediction set size when using or not using the ECDF correction.

3.5 Summary

We have presented two Monte Carlo CP techniques, Algorithms 1 and 2, which rely on sampling multiple synthetic pseudo-labels from $\mathbb{P}_{\text{agg}}^{Y|X}$ for each calibration data. In this setting, the coverage guarantees we

Algorithm 2 ECDF Monte Carlo CP with $(1-\alpha)(1-\delta)$ coverage guarantee.

Input: Calibration examples $(X_i, \lambda_i)_{i\in[n]}$; test example X; confidence levels α, δ; data split $1 \leq l \leq n-1$; number of samples m

Output: Prediction set $C(X)$ for test example X

1. Sample m labels $(Y_i^j)_{j\in[m]}$ per calibration example $(X_i)_{i\in[l]}$ where $\mathbb{P}(Y_i^j = k) = \lambda_{ik}$.
2. Sample one label Y_i per calibration example $(X_i)_{i\in\{l+1,\ldots,n\}}$ where $\mathbb{P}(Y_i = k) = \lambda_{ik}$.
3. Compute $(\bar{\rho}^i)_{i\in\{l+1,\ldots,n\}}$ where
$$\bar{\rho}^i = \frac{\sum_{j=1}^m \left(\sum_{p=1}^l \mathbb{I}[E(X_p, Y_p^j) \leq E(X_i, Y_i)] + 1\right)}{m(l+1)}. \tag{16}$$
4. Build the ECDF $\bar{F}(f) = \frac{1}{n-l}\sum_{i=l+1}^n \mathbb{I}[\bar{\rho}^i \leq f]$ and its upper bound $\bar{F}_+(f)$ using Equation (15).
5. For test example X, compute for $k \in [K]$
$$\bar{\rho}_k = \frac{\sum_{j=1}^m \left(\sum_{p=1}^l \mathbb{I}[E(X_p, Y_p^j) \leq E(X, k)] + 1\right)}{m(l+1)}, \quad \bar{\rho}_k^{\text{corr}} = \bar{F}_+(\bar{\rho}_k). \tag{17}$$
6. Return $C(X) = \{k \in [K] : \bar{\rho}_k^{\text{corr}} > \alpha\}$.

obtain have to be understood marginally across calibration examples *and* their sampled labels. We summarize the advantages and drawbacks of these methods in Table 1. In the following, we show that these developments are relevant beyond the setting of ambiguous ground truth and discuss related work.

Procedure	Theoretical coverage	Avg. empirical coverage	Variability emp. coverage	Additional calibration split
Alg. 1, $m = 1$	$1-\alpha$	$1-\alpha$	High	No
Alg. 1, $m \geq 2$	$1-2\alpha$	$1-\alpha$	Low	No
Alg. 2	$(1-\alpha)(1-\delta)$	$(1-\alpha)(1-\delta)$	Low	Yes

Table 1: Overview of the proposed Monte Carlo CP procedures in terms of the provided theoretical coverage guarantee, the observed empirical coverage (averaged across sampled pseudo labels), variability of the empirical coverage (w.r.t. sampled pseudo labels) and whether an additional split of the calibration data is required.

3.6 Applications to related problems

Multi-label classification: Let $\mathcal{Y}_i \subseteq [K]$ be the known, ground truth multi-label *set* for each calibration example X_i. For simplicity, we express the multi-label setting using plausibilities λ_i that divide probability mass equally across the $L_i = |\mathcal{Y}_i| \geq 1$ labels. We can then apply Monte Carlo CP to this setup. When $m \gg 1$, we obtain calibration examples where the proportion of $(Y_i^j)_{j\in[m]}$ equal to a given class is approximately $1/L_i$. This is related to an empirical existing method to perform multi-label CP (Tsoumakas & Katakis, 2007; Wang et al., 2014; 2015) where for each calibration example $(X_i, \mathcal{Y}_i)$ we perform CP using the calibration examples $(X_i, Y_i^1), \ldots, (X_i, Y_i^{L_i})$. Monte Carlo CP provides coverage guarantees on $\mathbb{P}_{\text{agg}}(Y \in C(X))$ for

$$\mathbb{P}_{\text{agg}}(Y = y | X = x) = \sum_{\mathcal{Y}} \lambda(\mathcal{Y})_y \; p(\mathcal{Y}|x) \tag{18}$$

with $p(\mathcal{Y}|x)$ being the conditional probability of the set $\mathcal{Y}$ given $X = x$ and $\lambda(\mathcal{Y})_y = \mathbb{I}(y \in \mathcal{Y})/|\mathcal{Y}|$.

Data augmentation and robustness: Consider a scenario where we have exchangeable calibration data $(X_i, Y_i)_{i\in[n]}$ for $Y_i \in [K]$. We want to augment the set of calibration data by using data augmentation, i.e. for each $X_i^1 := X_i$ we sample additional $X_i^2, ..., X_i^m \sim p(\cdot|X_i)$. For example, these could correspond to versions of X_i which are (adversarially or randomly) perturbed or corrupted, rotated, flipped, etc. As we usually train with data augmentation and frequently want to improve robustness against distribution shifts or specific types of perturbations, considering these augmentations for calibration is desirable. However, when using the augmented set $m \cdot n$ of calibration data $(X_i^j, Y_i)_{i\in[n],j\in[m]}$, the joint distribution of calibration data and test data is not exchangeable anymore. However, we can still provide rigorous coverage guarantees using a procedure very similar to Monte Carlo CP. For each test data (X, Y), we set $X^1 = X$ and sample augmentations $X^2, ..., X^m \sim p(\cdot|X)$. We then proceed by averaging the following m p-values

$$\rho_Y^j = \frac{\sum_{i=1}^n \mathbb{I}[E(X_i^j, Y_i) \leq E(X^j, Y)] + 1}{n+1}, \quad \bar{\rho}_Y = \frac{1}{m}\sum_{j=1}^m \rho_Y^j. \tag{19}$$

By following arguments similar to Section 3.3, the prediction set $C(X^1, X^2, ..., X^n) = \{y \in [K] : \bar{\rho}_y > \alpha\}$ satisfies $\mathbb{P}_{\text{aug}}(Y \in C(X^1, X^2, ..., X^n)) \geq 1 - 2\alpha$ where $\mathbb{P}_{\text{aug}}$ is the joint distribution of the test data $(X = X^1, Y)$ and the augmentations $(X^2, .., X^m)$. In this case, the coverage guarantee is marginal across $(X^1, ..., X^2, Y)$ and $(X_i^j, Y_i)_{i\in[n],j\in[m]}$.

3.7 Related work

CP (Vovk et al., 2005) has recently found numerous applications in machine learning, see e.g. (Romano et al., 2019; Sadinle et al., 2019; Romano et al., 2020; Angelopoulos et al., 2021; Stutz et al., 2021; Fisch et al., 2022). In this paper, we focus on split CP (Papadopoulos et al., 2002). However, there are also transductive and cross-validation/bagging-inspired variants being studied (Vovk et al., 2005; Vovk, 2015; Steinberger & Leeb, 2016; Barber et al., 2021; Linusson et al., 2020). Our work is related to these approaches in that many of them guarantee coverage $1 - 2\alpha$ while empirically obtaining coverage close to $1 - \alpha$. For example, cross-CP (Vovk, 2015) was recently shown to satisfy a $1 - 2\alpha$ guarantee in (Vovk et al., 2018; Kim et al., 2020). Moreover, this guarantee is also based on combining p-values without making any assumption about their dependence structure.

As outlined before, our work is also related to CP for multi-label classification which faces similar challenges as CP for ambiguous ground truth (Wang et al., 2014; 2015; Lambrou & Papadopoulos, 2016; Papadopoulos, 2014; Cauchois et al., 2021). Finally, our work has similarities to work on adversarially robust CP (Gendler et al., 2022), especially in terms of our ideal coverage guarantee.

There is also a long history of work on combining dependent or independent p-values (Fisher, 1925; Rüschendorf, 1982; Meng, 1994; Heard & Rubin-Delanchy, 2017). Key work has been done in (Vovk et al., 2018), showing results without dependence assumption and thereby establishing guarantees for, e.g., cross-CP. Similar to us, (Balasubramanian et al., 2015; Linusson et al., 2017; Toccaceli & Gammerman, 2019; Toccaceli, 2019) use the ECDF to combine p-values but they do not provide rigorous coverage guarantees for this procedure.

4 Applications

4.1 Main case study: skin condition classification

In the main case study of this paper, we follow (Liu et al., 2020; Stutz et al., 2023) and consider a very ambiguous as well as safety-critical application in dermatology: skin condition classification from multiple images. We use the dataset of Liu et al. (2020) consisting of 1949 test examples and 419 classes with up to 6 color images resized to 448×448 pixels. The classes, i.e., conditions, were annotated by various dermatologists who provide partial rankings. These rankings are aggregated deterministically to obtain the plausibilities λ using the inverse rank normalization procedure of (Liu et al., 2020) described in Section 3.1. We followed (Roy et al., 2022; Stutz et al., 2023) to train a classifier that achieves 72.6% top-3 accuracy